

Topped Up, Not Touched

An Instrument-Validation Study of Hand-Sanitiser Dispenser Refill Logs Against Covert Dispense Counts

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Background

Every hallway dispenser carries a small Facilities sticker logging its last refill as a hygiene-compliance data point. Does a full dispenser say anything about how often anyone actually used it, or only about how often someone remembered to check?

Aims

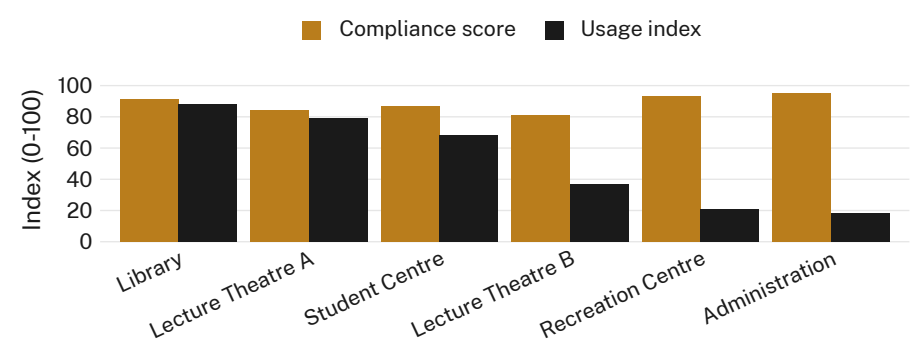
- Test whether refill-log frequency tracks actual dispense counts at the same units
- Compare covert actuation data against Facilities' published building-level compliance score
- Check whether either measure moves with illness-related staff and student absence

Methods

- 24 dispensers · 6 buildings · covert internal-actuation counters · 14 weeks
- Facilities refill-log timestamps and building compliance score pulled in parallel over the same window
- restocking held to the existing schedule; counter data withheld from Facilities throughout

Results

Refill frequency and covert actuation counts show weak agreement across buildings ($r = 0.21$, $n = 24$ dispensers). Illness-related absence tracked neither measure reliably over the term.



The administration building's dispensers were refilled on schedule throughout the term (13 visits between 2 March and 8 June 2026) while the covert counters logged a Usage Index of 18 — the lowest of the six buildings audited.

Discussion

A refill visit is easy to audit and hard to fake; whether anyone used the dispenser in between is a separate question the sticker was never built to answer. The building-level compliance score built from refill logs may track maintenance diligence more closely than hand hygiene. A well-serviced dispenser and a well-used one are not the same claim, and the current score has no way of telling them apart.

Next: extend the counters through a full academic year and add a self-report arm.

References

1. Marra, A. R., & Edmond, M. B. (2012). Automated hand hygiene count devices may better measure compliance than human observation. *Am. J. Infect. Control* 40(10), 955–959. doi:10.1016/j.ajic.2012.01.026
2. Srigley, J. A., Furness, C. D., Baker, G. R., & Gardam, M. (2014). Quantification of the Hawthorne effect in hand hygiene compliance monitoring using an electronic monitoring system. *BMJ Qual. Saf.* 23(12), 974–980. doi:10.1136/bmjqs-2014-003080
3. Boyce, J. M. (2017). Electronic monitoring in combination with direct observation as a means to significantly improve hand hygiene compliance. *Am. J. Infect. Control* 45(5), 528–535. doi:10.1016/j.ajic.2016.11.029
4. Gould, D. J., Lindström, H., Purssell, E., & Wigglesworth, N. (2020). Electronic hand hygiene monitoring: accuracy, impact on the Hawthorne effect and efficiency. *J. Infect. Prev.* 21, 136–143. doi:10.1177/1757177420907999
5. Wang, C., Jiang, W., Yang, K., Yu, D., Newn, J., Sarsenbayeva, Z., Goncalves, J., & Kostakos, V. (2021). Electronic monitoring systems for hand hygiene: systematic review of technology. *J. Med. Internet Res.* 23(11), e27880. doi:10.2196/27880

Ethics approval 2026/118; actuation counts aggregated to building level, no individual or CCTV data collected; thresholds pre-registered.

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